

TASK 1 — LINUX IP INVESTIGATION

```
ec2-user@jayeshpatil:~$ ip a
[ec2-user@jayeshpatil ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state
    UP group default qlen 1000
    link/ether 0a:ff:d8:85:da:21 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope globa
1 dynamic ens5
    valid_lft 3305sec preferred_lft 3305sec
    inet6 fe80::8ff:d8ff:fe85:da21/64 scope link proto kernel_l1
    valid_lft forever preferred_lft forever
[ec2-user@jayeshpatil ~]$
```

Screenshot 1 — IP address and interface

```
ec2-user@jayeshpatil:~$ ip -4 addr
[ec2-user@jayeshpatil ~]$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qd
isc mq state UP group default qlen 1000
    altname enp0s5
    altname eni-001822ae9b0ac9ab1
    altname device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255
    scope global dynamic ens5
        valid_lft 2974sec preferred_lft 2974sec
[ec2-user@jayeshpatil ~]$ hostname
jayeshpatil
[ec2-user@jayeshpatil ~]$ hostname -I
172.31.16.110
[ec2-user@jayeshpatil ~]$ ip -4 route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.110 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.16.110 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.16.110 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.16.110 metric 512
[ec2-user@jayeshpatil ~]$ curl ifconfig.me
54.92.222.22[ec2-user@jayeshpatil ~]$ curl ifconfig.me
54.92.222.22[ec2-user@jayeshpatil ~]$ ping -c 4 192.168.1.1
ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
```

Screenshot 2 — IPv4 address, interface and route

```
ec2-user@jayeshpatil:~  
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.16.110  
metric 512  
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.16.110 metri  
c 512  
[ec2-user@jayeshpatil ~]$ cat /etc/resolv.conf  
# This is /run/systemd/resolve/resolv.conf managed by man:systemd-  
resolved(8).  
# Do not edit.  
#  
# This file might be symlinked as /etc/resolv.conf. If you're look  
ing at  
# /etc/resolv.conf and seeing this text, you have followed the sym  
link.  
#  
# This is a dynamic resolv.conf file for connecting local clients  
directly to  
# all known uplink DNS servers. This file lists all configured sea  
rch domains.  
#  
# Third party programs should typically not access this file direc  
tly, but only  
# through the symlink at /etc/resolv.conf. To manage man:resolv.co  
nf(5) in a  
# different way, replace this symlink by a static file or a differ  
ent symlink.  
#  
# See man:systemd-resolved.service(8) for details about the suppor  
ted modes of  
# operation for /etc/resolv.conf.  
nameserver 172.31.0.2  
search ec2.internal  
[ec2-user@jayeshpatil ~]$ hostname -I  
172.31.16.110  
[ec2-user@jayeshpatil ~]$
```

Screenshot 3 — DNS configuration

TASK 2 — IPv4 ADDRESS ANALYSIS

```
ec2-user@jayeshpatil:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    altnam enp0s5
    altnam eni-001822ae9b0ac9ab1
    altnam device-number-0.0
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3429sec preferred_lft 3429sec
ec2-user@jayeshpatil ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=1.05 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=1.05 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=1.07 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=1.07 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.048/1.060/1.073/0.010 ms
ec2-user@jayeshpatil ~]$ ping -c 4 google.com
PING google.com (172.253.115.101) 56(84) bytes of data:
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=1 ttl=106 time=1.49 m
s
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=2 ttl=106 time=1.55 m
s
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=3 ttl=106 time=1.54 m
s
64 bytes from bg-in-f101.1e100.net (172.253.115.101): icmp_seq=4 ttl=106 time=1.59 m
s

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.488/1.542/1.589/0.035 ms
ec2-user@jayeshpatil ~]$
```

Screenshot 4 — IPv4 and network configuration

The screenshot displays the AWS Management Console interface. On the left, the navigation menu shows 'EC2' selected under 'Instances'. The main content area is titled 'Instances (1/9)' and shows a table of instances. The instance 'volume2' with ID 'i-099754dfd4215f2c6' is highlighted, showing a state of 'Running' and 'Initializing'. Below the table, the details for this instance are shown, including a table of security group rules. The rules table has columns for Name, Security group rule ID, Port range, Protocol, and Source. Two rules are listed: one for port 22 (TCP) and one for port 80 (TCP), both with a source of '0.0.0.0/0'.

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-0508d45238794649b	22	TCP	0.0.0.0/0
-	sgr-0e7149e0a87ff61e9	80	TCP	0.0.0.0/0

Screenshot 5 — Connectivity test

TASK 3 — DYNAMIC IP INVESTIGATION

```
ec2-user@jayeshpatil:~  
Admin@jaypc MINGW64 ~/downloads  
$ ssh -i linux.pem ec2-user@54.92.222.22  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
  
#####  
#####  
#####  
#/  
V-+>  
->  
-/m/<
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

Last login: Fri Sep 11 11:40:16 2026 from 103.252.53.70
[ec2-user@jayeshpatil ~]\$

Screenshot 6 — Current EC2 connection

The screenshot displays the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, search bar, and user information (jaysheP). The main header shows "EC2 > Instances". On the left, a sidebar lists navigation options like Dashboard, AWS Global View, Events, and various instance-related services. The central pane shows a list of EC2 instances under the heading "Instances (1/9)". A table lists three instances: "volume2" (Running), "volume" (Stopped), and "portfolio" (Stopped). Below the table, the details for instance "i-099754dfd4215f2c6" are expanded, showing tabs for Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags. The "Details" tab is active, displaying the Instance ID, Public IPv4 address (54.92.222.22), Private IPv4 addresses (172.31.16.110), and other metadata.

```
ec2-user@jayeshpatil:~  
Admin@jaypc MINGW64 ~/Downloads  
$ ssh -i linux.pem ec2-user@54.92.222.22  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
  
#_      _____  
~\     #####  
~~~   \#####\      Amazon Linux 2023  
~~~~  \###|  
~~~~  \|#/    https://aws.amazon.com/linux/amazon-linux-2023  
~~~~  V~'____->  
~~~~  
~~~~  _-_-_  
~~~~ /'_/_/_/_/  
~~~~/_m/'
```

Last login: Fri Sep 11 11:42:03 2026 from 103.252.53.70
[ec2-user@jayeshpatil ~]\$ curl -4 ifconfig.me
54.92.222.22[ec2-user@jayeshpatil ~]\$ |

Screenshot 8 — SSH connection and public IP

TASK 5 — CLOUD NETWORK TROUBLESHOOTING

```
ec2-user@jayeshpatil:~  
[ec2-user@jayeshpatil ~]$ sudo systemctl status nginx  
● nginx.service - The nginx HTTP and reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)  
   Active: active (running) since Fri 2026-09-11 11:51:03 UTC; 9s ago  
     Process: 3778 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)  
     Process: 3779 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)  
     Process: 3780 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)  
    Main PID: 3781 (nginx)  
       Tasks: 3 (limit: 1059)  
      Memory: 3.6M  
         CPU: 58ms  
    CGroup: /system.slice/nginx.service  
            └─3781 "nginx: master process /usr/sbin/nginx"  
              └─3782 "nginx: worker process"  
                └─3783 "nginx: worker process"  
  
Sep 11 11:51:03 jayeshpatil systemd[1]: Starting nginx.service - The nginx HTTP and  
Sep 11 11:51:03 jayeshpatil nginx[3779]: nginx: the configuration file /etc/nginx/nginx  
Sep 11 11:51:03 jayeshpatil nginx[3779]: nginx: configuration file /etc/nginx/nginx  
Sep 11 11:51:03 jayeshpatil systemd[1]: Started nginx.service - The nginx HTTP and  
  
[ec2-user@jayeshpatil ~]$ ip -4 addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default ql  
en 1000  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default  
qlen 1000  
    altname enp0s5  
    altname eni-001822ae9b0ac9ab1  
    altname device-number-0.0  
    inet 172.31.16.110/20 metric 512 brd 172.31.31.255 scope global dynamic ens5  
        valid_lft 3429sec preferred_lft 3429sec  
[ec2-user@jayeshpatil ~]$ ping -c 4 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=1.05 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=1.05 ms
```

Screenshot 9 — Nginx service status

```
ec2-user@jayeshpatil:~  
[ec2-user@jayeshpatil ~]$ sudo ss -tulpn  
Netid State Recv-Q Send-Q           Local Address:  Peer Address:Port  
Process  
udp    UNCONN 0      0           0.0.0.0:*        127.0.0.1:  
323    users:(("chronyd",pid=1896,fd=5))  
udp    UNCONN 0      0           172.31.16.110%ens5: 0.0.0.0:*  
68     users:(("systemd-network",pid=1708,fd=17))  
udp    UNCONN 0      0           [::]:*          [::]:1  
323    users:(("chronyd",pid=1896,fd=6))  
udp    UNCONN 0      0           [fe80::8ff:d8ff:fe85:da21]%ens5: [::]:*  
546    users:(("systemd-network",pid=1708,fd=19))  
tcp    LISTEN 0      511         0.0.0.0:*        0.0.0.0:  
80     users:(("nginx",pid=3783,fd=8),("nginx",pid=3782,fd=8),("ngin  
x",pid=3781,fd=8))  
tcp    LISTEN 0      128         0.0.0.0:*        0.0.0.0:  
22     users:(("sshd",pid=1868,fd=7))  
tcp    LISTEN 0      511         [::]:*          [::]:  
80     users:(("nginx",pid=3783,fd=9),("nginx",pid=3782,fd=9),("ngin  
x",pid=3781,fd=9))  
tcp    LISTEN 0      128         [::]:*          [::]:  
22     users:(("sshd",pid=1868,fd=8))  
[ec2-user@jayeshpatil ~]$  
[ec2-user@jayeshpatil ~]$ curl http://localhost  
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Welcome to nginx!</title>  
    <style>  
      html { color-scheme: light dark; }  
      body { width: 35em; margin: 0 auto;  
font-family: Tahoma, Verdana, Arial, sans-serif; }  
    </style>  
  </head>  
  <body>  
    <h1>Welcome to nginx!</h1>  
    <p>If you see this page, nginx is successfully installed and w  
orking.  
    Further configuration is required for the web server, reverse  
proxy,  
API gateway, load balancer, content cache, or other features.<
```

Screenshot 10 — Listening ports